

Sea-ice sub-cycling in the FESOM2 GPU port: what pays, and what it costs

N. Koldunov, 6 August 2026 — summary of the mEVP communication study

Three ways to stop exchanging the halo 120 times per step

mEVP runs 120 sub-cycles per model time step and currently exchanges the ice-velocity halo after every one. Three ways to do it less often were implemented and measured:

delayed exchange — exchange every K -th sub-cycle and let the halo go stale. Fastest by far, *but it writes the sub-domain boundaries into the ice field: not usable* (below). **Wide halo** (Danilov) — exchange every K -th sub-cycle but recompute a K -deep ghost ring in between, so nothing is stale: *exact*, bit for bit, and no imprint; smaller gain. **Divergence form** (Danilov) — carry $\nabla \cdot \sigma$ at nodes instead of σ at elements: exact to 10^{-15} , pays on CPU, costs on GPU.

What they buy — in both currencies

Percent of the *model step* is what a production run feels; percent of the *ice cost* (ice + ice dynamics + ice advection, compute *and* MPI wait) is what judges a sea-ice scheme. They differ 5–10 $\times$, and quoting only the first makes a good scheme look like nothing.

		GPU		CPU, at scale	
		ice cost	model step	ice cost	model step
delayed exchange	$K=4$	–37 to –58%	–8.5 to –14%	–5 to –25%	–0.3 to –3.6%
wide halo, exact	$K=8$	–12 to –19%	–2 to –7%	never wins	
divergence form alone	—	+2 to +17%	0 to +1.6%	–2 to –8.5%	–0.5 to 0%

The CPU columns grow with core count (–5% of the ice at 128 ranks, –25% at 2048); the GPU columns shrink with the mesh a GPU holds (–58% at 32k nodes per GPU, –37% at 460k). The divergence form is **not** a null result: on CPU at 512 ranks and above it takes 2–8.5% off the ice and is simply invisible in the model step. On GPU it costs, because there the sub-cycle is limited by the number of GPU operations issued rather than by arithmetic or memory traffic, so removing element arrays removes work that was not being paid for. Danilov’s own $\sim 20\%$ for the divergence form with its double halo matches our –19.1% **of the ice cost** for that combination (fArc, 16 GPU nodes).

Why the fast option is out: it draws the decomposition into the ice

The delayed exchange reads halo values up to $K - 1$ sub-cycles old, and those nodes are exactly the one-cell rim of each sub-domain. The partition is static, so the same nodes are perturbed every step of the run. Measured on fArc, 16 GPUs, 60 days of daily output — mean $|\Delta \text{ ice thickness}|$ on the rim against the interior:

	rim	interior	ratio
two identical runs (control — no lever at all)	0.00105 m	0.00124 m	0.85
wide halo, exact, $K=8$	0.00104	0.00120	0.87
delayed exchange, $K=2$	0.00432	0.00249	1.70
delayed exchange, $K=4$	0.01178	0.00452	2.52

It is there on **day one** (ratio 3.1), persists for two months, and is visible as a web of differences following the sub-domain boundaries. Reducing the delay to $K=2$ halves it but does not remove it. The exact wide halo scores like the control, as it must, which also validates the diagnostic. **An artificial polygonal structure in the ice, in the shape of the processor grid, is not something a sea-ice model can carry** — and it is a different objection from the accuracy cost below, which is small. The ring profiles and the map are Figure 4 of the full note.

What the delayed exchange costs in simulated climate (1 year, CORE2, exactly reproducible)

Pattern correlation against the same run exchanging every sub-cycle, judged against the difference between two independent correct implementations of mEVP (our C port versus Fortran FESOM2). At $K=4$: ice concentration 0.999995 and thickness 0.999994 against a yardstick of 0.99999/0.99998; ice velocity 0.963/0.950 against 0.954/0.939. Every field stays inside the yardstick at $K \leq 8$, but only by 1.1–1.5 $\times$ on ice velocity. Hemispheric ice extent moves 0.23% of its seasonal amplitude and the ocean surface is unchanged to six digits. *CORE2 resolution only.*

Recommendation

For production, the exact wide halo. Bit-for-bit identical to the standard scheme, so no accuracy argument and no partition imprint, and worth 12–19% of the ice cost on GPU at $K=8$. It does not pay on CPU on any mesh.

The delayed exchange is a no-go, despite being worth three times as much, and we report it as a measured negative: it is the fastest thing we found and it cannot be used, because it puts the processor decomposition into the ice field. No amount of speed buys that back.

The divergence form is worth having on CPU at scale and is a cost on GPU — and it is what makes the wide halo cheapest to carry.